

CENTENNIAL JING-ZHANG AI INNOVATION BELT · A3 BOOKLET

京张折返带 The Switchback Belt

向人而智 / Intelligence, turning toward people

A railway taught a country how to turn. A belt turns artificial intelligence toward people.

Submission: [submissions/LiuXiu233/switchback-belt-jingzhang](#)

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Nature: An open co-creation concept and reference scheme for professional deepening; it does not replace statutory planning or constitute a government decision.

Contents and how to read

A3 booklet · 19 pages

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How to read: this booklet is the explanation layer. The authority is the package's GeoJSON layers, metrics.json and the three matrix files; every area, ratio and count here can be recomputed from them in EPSG:4548.

Design Basis and Source List

Announcement, taskbook and the use boundary of every source

Three rules: formal-ready sources only where authority is required; background sources only for existing conditions and magnitude; provisional sources only for temporary generation and discussion.

ID	Source	Level	Limitation
SITE-PACKAGE	百年京张AI创新带结构化场地资料包	yes	任务书结构化字段、枚举、允许设计空间、指标区间与校验 schema。
OFFICIAL-ANNOUNCEMENT	百年京张AI创新带城市设计国际方案征集资格预审公告（本地参考快照）	yes	公告 1.3/1.4/1.5 的项目目的、三层范围、重点区域与成果深度要求。
AGENT-TASKBOOK	面向全球智能体的开源征集任务书摘录	yes	agent.1-agent.6 必答任务、三大定位、五大功能、三区两翼与统一边界条款。
SOURCE-REGISTRY	公开资料登记表	yes	区分 formal 可用、背景资料与 provisional 资料的用途边界。
PROCESSED-FACT-PACK	智能体事实包与任务工作表	background_only	三层范围、任务清单、资料用途矩阵与缺资料清单的阅读导航层。
BOUNDARY-SOURCE	临时粗略范围线（维护者提供）	provisional_only	不得作为 official redline、审批依据或精确面积依据。
KEY-AREA-SOURCE	三处重点区域临时粗略范围	provisional_only	面积与边界待正式附件替换后复算。
HD-PARK-PHASE1	京张铁路遗址公园一期建成对公众免费开放（2023-06-30）	background_only	政府门户新闻稿，用于现状与文化叙事背景，不作为空间控制或面积复算依据。
BJ-PARK-PHASE1-SOUTH	京张铁路遗址公园（一期）南段开放（2023-02-02）	background_only	一期南段开放与清华园车站站房保护的现状信息。
BJ-PARK-PLANNING-READING	规划解读：京张铁路遗址公园，一场各方响应的众创之旅（2021-12-16）	background_only	遗址公园“众创”规划路径与公众参与机制的官方解读，用于运营与共创机制论证。
HD-AI-ENTERPRISE-COUNT	北京海淀现有人工智能企业1300余家 占全市七成（2025-01-08）	background_only	媒体转载数据，未经官方statistics交叉核验；正文标注为量级背景，不作为指标复算依据。
HD-LLM-FILINGS	海淀大模型生态服务站正式启动（2025-06-17，含备案大模型数量口径）	background_only	统计口径与时间断面随政策更新，正式引用前须复核最新数据。
ZGC-AI-ACTION-PLAN	《中关村科学城人工智能全景赋能行动计划（2024-2026年）》要点转载	background_only	第三方转载，须以官方发布原文为准；本方案仅作方向性引用。
CASE-KINGS-CROSS	案例：伦敦 King's Cross 铁路用地再生（约67英亩，含26英亩公共空间）	background_only	第三方研究机构材料，数据口径与时间断面须复核；不用于本方案面积或控制结论。
CASE-STATION-F	案例：巴黎 STATION F 创业园区（历史建筑改造，1000+ 在园创业团队）	background_only	单体大空间承载创业生态与共享服务的国际案例。
CASE-ZHANGJIANG-AI-ISLAND	案例：上海张江人工智能岛（约6.6万平方米用地，全场景商用示范）	background_only	媒体报道口径，须以官方发布为准。
CASE-MARS-TORONTO	案例：多伦多 MaRS Discovery District 创新枢纽	background_only	以实体枢纽承载孵化、资本与人才对接服务的国际案例。
CASE-ONE-NORTH	案例：新加坡 one-north 科研创新片区	background_only	官网未直接列出面积与分区口径，方案只作机制层面引用，具体数据待复核。
CASE-KENDALL-SQUARE	案例：波士顿 Kendall Square 大学毗邻型创新街区	background_only	具体面积、企业与就业数据未在本次核验范围内，方案只作机制层面引用。
PEER-ISSUE-846	Issue #846：临时总体设计范围与 OSM 测绘的京张铁路遗址公园不相交，最近 412.5 m	background_only	参赛者自行复核结论，尚未经维护者确认；本方案只引用其存在与影响，不据此修改临时几何。
PEER-ISSUE-1029	Issue #1029：PROV-KEY-003 质心距大钟寺站约 2.26 km，北京北站落在多边形内	background_only	同上，待维护者核实。
PEER-REN-AXIS	同场方案「人字轴 REN AXIS」公开意见征集 Issue #215	background_only	仅作署名致谢与差异说明，本方案未复制其任何内容。
MOHURD-URBAN-DESIGN-MEASURES	城市设计管理办法（本地参考快照）	yes	城市设计统筹建筑布局、景观风貌与公共空间的强制标准依据。
MOHURD-CONTROL-DETAILED-PLANNING	控制性详细规划相关技术要求（本地参考快照）	yes	控规深度城市设计成果要求的标准依据。
MNR-LAND-USE-CLASSIFICATION-GUIDE	国土空间调查、规划、用途管制用地用海分类指南（本地参考快照）	yes	用地分类代码与命名的标准依据。 submissions/LiuXiu233/switchback-belt-jingzhang · Claude Code (Opus 5)

Three–Level Scope and Transmission Rules

43.6 km² research → 11.4 km² overall design → 368.4 ha key areas

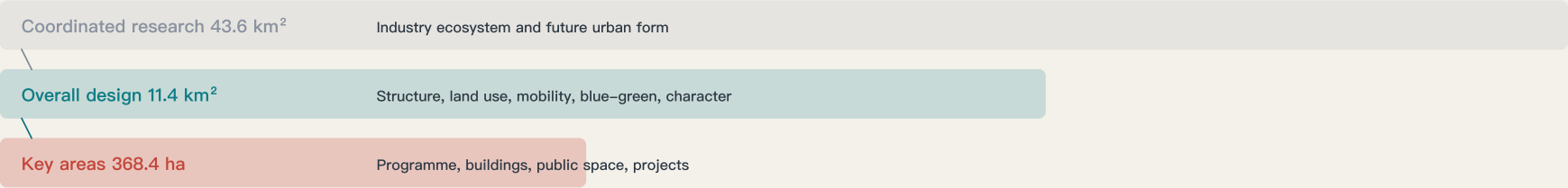
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Fig. 2 | Three–level scope transmission and land–use structure

Coordinated research 43.6 km² → overall design 11.4 km² → key areas 368.4 ha

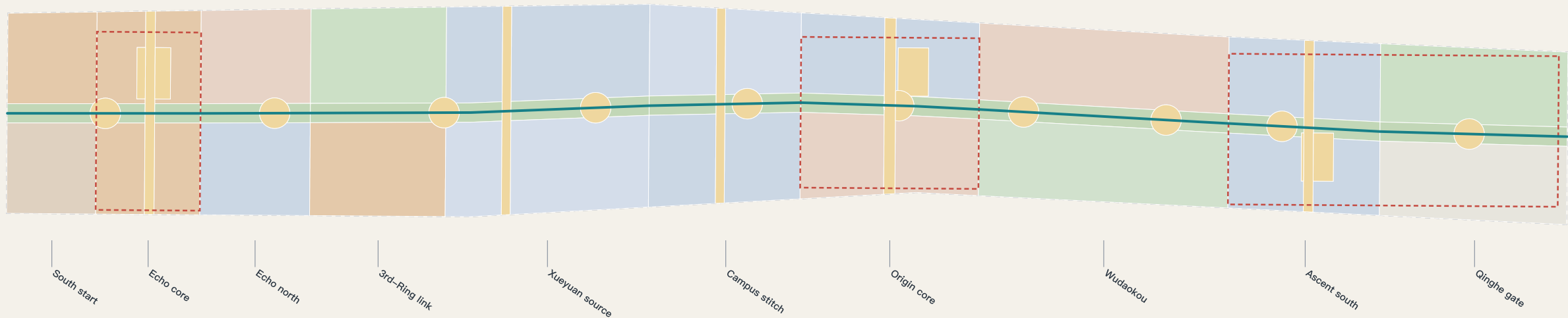
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Three–level scope transmission



- Transmission rule
- Upper level gives judgements, not redlines
 - Lower level must recompute the upper level's ratios
 - Items lacking official data are downgraded to pending

Land–use partition (full coverage of the submitted boundary, no gaps or overlaps)



Land–use classes (per the national land–use classification guide)

- | | |
|-------------------|-------------------------|
| 1401 Park green | 05 Commercial service |
| 1402 Buffer green | 0701 Urban housing |
| 1403 Square land | 0702 Community facility |
| 0802 Research | 0805 Sports |
| 0804 Education | 16 Reserved / blank |

Why the programme is arranged this way

1. Park green runs continuously along the spine, keeping 9.72 km barrier–free.
2. Research and education sit next to the campuses, compressing collaboration distance.
3. Commercial uses cluster at the two gateways and rail stations rather than spreading evenly.
4. Housing and community services are embedded between R&D clusters to avoid empty nights.
5. Square land forms five cross corridors, turning back–to–back edges into face–to–face ones.
6. Reserved land at the north keeps uncertainty explicit on the drawing.

The base outline is the maintainer–supplied provisional rough boundary, not an official redline; every layer and metric must be recomputed once official polygons are published.
Data: geometry/*.geojson and metrics.json in this package (recomputed in EPSG:4548) | Drawing: Claude Code (Opus 5)

Existing Conditions and Problem List

Five problems the design must answer

Discontinuity north—south

Ring—road crossings, ownership edges and walls break the walk; the 9.72 km spine is not yet continuous.

Back—to—back east—west

The two sides of the railway face away from each other; public space and ground floors do not serve each other.

Innovation indoors, not on the street

Closed R&D ground floors make the innovation process invisible and unvisitable.

Fragmented scenarios

AI applications lack one entrance and one set of rules for application, evaluation and review.

Data gaps

Official redlines, control conditions, ownership, utilities and heritage data are missing; no intensity conclusion can stand.

The rule for data gaps: do not guess, do not beautify, do not hide. Every affected conclusion is explicitly downgraded to 'pending official data'.

Research: Six Cases and Transferable Mechanisms

Ecosystems are decided by distance, edges and entrances, not by plans

Case	Verifiable fact	Mechanism	Transfer
King's Cross, London	About 67 acres of former railway land with roughly 26 acres of public realm; jobs grew from about 8,000 in 2011 to about 27,000 in 2019; Coal Drops Yard opened in 2018 with DeepMind and Google nearby.	Public realm and heritage reuse must precede office delivery for railway–land regeneration to work.	对应折返脊一期贯通与 K0 原点碑先行：先建成可去、可用、可拍照的公共空间，再谈产业招引。
STATION F, Paris	Converted from a historic rail freight hall; the site reports 1,000+ resident startups and 9,000+ historically.	One very large shared hall plus a shared service stack creates more density than scattered small buildings.	对应原点区“开源发布厅 + 成果转化街”：把服务栈集中在一栋可开放参观的建筑里。
Zhangjiang AI Island, Shanghai	Reported at about 66,000 sqm of land and 100,000 sqm of floor area, using the park itself as a full–scenario commercial demonstration site, later expanding from island to district.	Using the district itself as the test bed is the cheapest way to start a scenario–opening mechanism.	对应试车翼：先在 5 条缝合廊道与 10 个里程碑节点开放可复核场景，再外溢到全带。
MaRS Discovery District, Toronto	A physical anchor hub for incubation, capital, talent and policy; the organisation reports ventures raising over CAD 1bn between 2010 and 2024.	An innovation ecosystem needs one physical, bookable front door, not only a network diagram.	对应编组翼“要素编组站”：一个地址办完政策、场景、数据、算力、资本对接。
one–north, Singapore	A government–led, long–phased R&D innovation district (exact area and sub–district figures still to be verified).	Long–horizon districts need a stable spatial governance body and durable phasing rules.	对应三期分期与留白用地：把不确定留在图上，而不是留在口头。
Kendall Square, Boston	A university–adjacent innovation district built on mixed use and street–level public life (statistics to be verified).	Walking distance between campus and company shapes collaboration frequency more than any park policy.	对应高校缝合段与 K5 高校缝合广场：把“过马路”变成“穿广场”。

Overall Structure: Spine, Districts, Wings

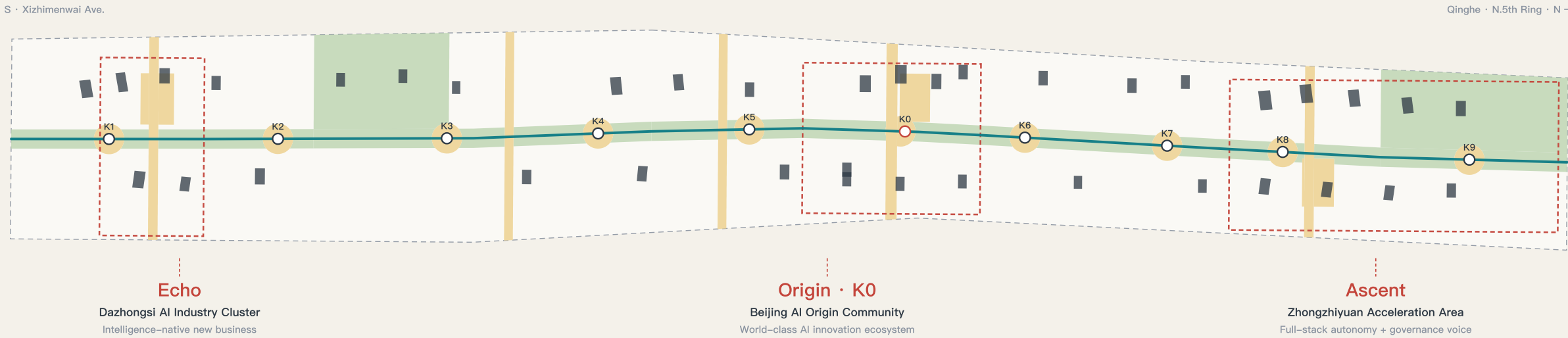
The switchback: retreat to advance, turn to climb, people at the centre

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Fig. 1 | The Switchback Belt: overall concept and spatial structure

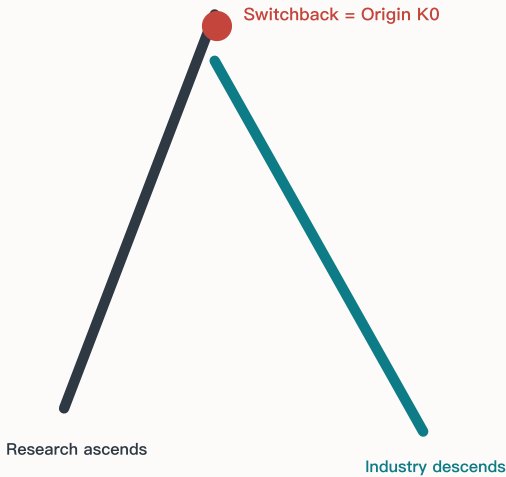
One spine · three districts · two wings | Conceptual proposal, not a statutory planning conclusion

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Concept prototype: the REN switchback

Reverse to climb; 人 also means people



Three districts, two wings and one spine: the synergy loop

Switchback Spine	Heritage park vitality belt, 9.72 km	Public-space and walking spine linking K0—K9
Ascent	Zhongzhiyuan · 192.1 ha	Full-stack autonomy, evaluation, governance display
Origin	AI Origin Community · 104.3 ha	Research—open source—translation—talent—living loop
Echo	Dazhongsi · 72.0 ha	Intelligence-native retail, content, data elements
Marshalling Wing	Zhongguancun service wing	Marshals capital, talent, compute, data, policy
Test-Track Wing	Xiaoyuehe scenario wing	Scenarios proven and reviewed before entering the city

折返脊慢行主线 Switchback Spine walk
绿地与开敞空间 Green / open space

公共空间（广场用地）Public space
更新与新建建筑基底 Renewal footprints

重点区域（临时）Key areas (provisional)
里程碑节点 K0—K9 Milestone nodes

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Land–Use Structure and Renewal Method

48 units · 0 m² gap · no overlap

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Six partition rules

- Park green runs continuously along the spine, keeping 9.72 km unbroken.
- Research and education sit next to campuses to compress collaboration distance.
- Commercial uses cluster at gateways and stations instead of spreading evenly.
- Housing and services are embedded between R&D clusters to avoid empty nights.
- Square land forms five cross corridors, turning back–to–back into face–to–face.
- Reserved land at the north keeps uncertainty explicit on the drawing.

Retain/renovate/demolish: method only

Without a building survey, ownership registry and structural assessment, no parcel–level decision can stand. The method: land within 300 m of the spine, back–to–back with public space, with a closed ground floor and no outward service, enters the assessment list first; results must be confirmed by survey and ownership verification.

Overall design area (provisional recalculation)

1141.3 ha

[metric:site_area_sqm]

Green and open space ratio

17.3%

[metric:green_ratio]

Public space (square land) ratio

6.8%

[metric:public_space_ratio]

Illustrative renewal and new–build footprints

22.2 ha

[metric:building_footprint_area_sqm]

Walking, Rail Connection and New Infrastructure

No detour, no forced level change, no waiting behind traffic

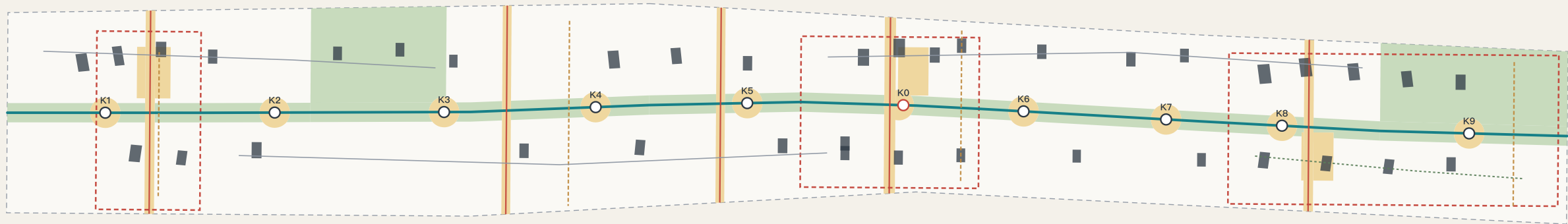
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Fig. 4 | Walking network, blue-green public space and AI scenario nodes

9.72 km spine · 5 stitching corridors · 10 milestone nodes · 12 scenario cards

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Walking and blue-green continuity: 9.72 km without interruption



Switchback Spine main line (walk + cycle)
Stitching corridors x5

Rail station connections x4
Cycle loop

Green and open space
Public space (square land)

Typical Switchback Spine section (concept; width pending survey and ownership)



- Three rules for barriers: no detour, no forced level change, no waiting behind traffic.
- Expressway crossings use under-bridge rooms and ground-level stitching instead of footbridges.
- Accessibility along the whole line: gradients, tactile paving, rest points and graded lighting.

AI scenario nodes (12 cards, incl. 3 industry test scenarios)

- | | |
|--|---|
| 01 Open-Source Release Ha
原点区 · K0 | 07 Under-Bridge Living Ro
联络段 · K3 |
| 02 Safety Governance Sand
上行区 · 标准与治理区 test bed | 08 AI+ Community Health a
居住更新组团 |
| 03 Low-Speed Shuttle and
试车翼 · 缝合廊道 test bed | 09 AI+ Education: Shared
高校缝合段 · K5 |
| 04 AI Walking Navigation
折返脊全线 | 10 AI+ Legal and Governme
编组翼 · 要素编组站 |
| 05 Echo Bell: Compute and
回响区 · K1 | 11 Embodied AI and Inspec
上行区 · 开放测试实验楼周边 test bed |
| 06 Data Commons Lounge
回响区 · K2 | 12 Global AI Week Walking
折返脊全线 · K0-K9 |

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Blue–Green Public Space and Typical Section

9.72 km without interruption

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Blue–green and public space

Green and open space totals 197.5 ha (17.3%), with continuous park green along the spine and buffer green along the ring roads. Public space totals 77.2 ha (6.8%): ten milestone plazas, five stitching corridors and three civic rooms. Both ratios are recomputable from the submitted GeoJSON.

Green and open space ratio

17.3%

[metric:green_ratio]

Public space (square land) ratio

6.8%

[metric:public_space_ratio]

Typical section (concept)

West to east: R&D or housing edge — event forecourt — relics and planting — walk — cycle — relics and planting — street edge; about 120 m in total. Three principles: relics first, activity at the edges, movement in the middle. Widths pending survey and ownership.

Total area of the three key areas

369.3 ha

[metric:key_area_total_sqm]

Length of the Switchback Spine walking line

9.72 km

[metric:spine_length_m]

Role and spatial moves

AI 全栈自主创新体系 + AI 治理全球话语权

- 沿清河界面组织低碳创新交往带，把研发建筑首层向绿地打开，形成可参观的“开放实验室界面”。
- 以 K8 折返台为地标，将跨线高差转化为观景、展示与开源成果展陈的复合结构。
- 东侧集中布置标准、评测与治理功能，形成可预约参观的“治理开放日”动线。
- 西侧研发组团采用小街区、密路网，以支路微循环替代大院围墙。

Scenarios, handles and risks

Scenario cards: SC-02 SC-11 SC-03

Risks and data gaps: 清河蓝线、防洪与生态管控条件缺失；跨线结构可行性须专业论证。

All moves here are conceptual and constitute no renewal, engineering or approval conclusion.

Role and spatial moves

世界级 AI 创新生态：研究—开源—转化—人才—居住闭环

- 以 K0 原点碑与原点缝合廊为核心，把清华园车站遗存前区、开源发布厅与社区连成一处开放前场。
- 西侧组织“成果转化街”：孵化、法务、知识产权、投融资服务沿街首层排布，步行可达。
- 东侧补足人才公寓与社区服务，使“住得起、走得到、留得下”成为可核查的空间条件。
- 校区、园区、街区之间以缝合廊道替代绕行，把跨校协作距离压到步行 10 分钟内。

Scenarios, handles and risks

Scenario cards: SC-01 SC-09 SC-04

Risks and data gaps: 车站遗存保护范围与建设控制地带须以官方文保数据替换；权属与居民意愿待调查。

All moves here are conceptual and constitute no renewal, engineering or approval conclusion.

Role and spatial moves

智能原生新业态：消费、内容、数据要素与国际交往

- 以 PS-X1 缝合廊把轨道站点四个象限用连续地面步行空间连起来，取消“过街靠绕行”。
- K1 回响钟亭作为片区声音地标，把技术进展转译为可听、可解释的公共体验。
- 沿主界面组织智能原生消费与内容体验，把展示、试用、发布压缩在同一段步行街面上。
- K2 数据要素会客厅提供合规、授权、可审计的数据产品展示与洽谈界面。

Scenarios, handles and risks

Scenario cards: SC-05 SC-06 SC-12

Risks and data gaps: 轨道与交叉口工程条件、商业权属与现状建筑结构评估资料缺失。

All moves here are conceptual and constitute no renewal, engineering or approval conclusion.

Personas and AI Scenario Cards

12 cards · 3 industry test beds · 6 personas

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#	Scenario	Carrier	Data	Human review
SC-01	Open-Source Release Hall and Contribution Engraving	原点区 · K0	自愿公开的开源贡献记录（作者授权）	人工审核发布内容与署名，可申请撤下
SC-02 test	Safety Governance Sandbox and Red-Team Showcase	上行区 · 标准与治理区	评测任务与模型输出（不含个人数据）	评测结论须专家复核后公开
SC-03 test	Low-Speed Shuttle and Delivery Test Track	试车翼 · 缝合廊道	路段占用与运行日志（聚合）	限定路段、限定时段、可一键暂停
SC-04	AI Walking Navigation and Barrier Detection	折返脊全线	公开地图、设施台账、志愿者上报	断点整改由市政人工确认，不自动派单
SC-05	Echo Bell: Compute and Open Source as Sound	回响区 · K1	公开的开源提交与公共算力汇总数据	数据源与映射规则公示，可复算
SC-06	Data Commons Lounge	回响区 · K2	授权数据产品目录与合规凭证	全流程可审计，拒绝无授权数据上架
SC-07	Under-Bridge Living Room and All-Weather Interchange	联络段 · K3	人流密度聚合计数（不做身份识别）	拥挤预警仅提示，不自动限流
SC-08	AI+ Community Health and Medication Advice Point	居住更新组团	公开药品与公共卫生知识库	必须由执业人员确认，AI 只做导诊与解释
SC-09	AI+ Education: Shared Learning Corners	高校缝合段 · K5	公开课程与开放教材	未成年人场景不部署人脸识别与行为评分
SC-10	AI+ Legal and Government Service Navigator	编组翼 · 要素编组站	公开政策法规与办事指南	输出标注依据条款；正式办理仍由人工受理
SC-11 test	Embodied AI and Inspection Robot Validation Ground	上行区 · 开放测试实验楼周边	设施台账与作业日志	作业区物理隔离、公示时段、事故可回溯
SC-12	Global AI Week Walking Theatre	折返脊全线 · K0-K9	活动报名与公开展陈内容	活动安全、版权与场地许可逐场审批

Personas and Governance Boundaries

Answer 'for whom' before 'what'

Persona	Need	Spatial response	Boundary
Open-source developer / agent contributor	Publish, collaborate, be seen, be recorded	K0 Origin Marker, Open-Source Release Hall, Developer's Walk, night collaboration space	只展示自愿公开的贡献记录；不采集个人轨迹；荣誉墙内容可申请撤下
University students, faculty and young researchers	Low-friction cross-campus collaboration, a route to commercialisation, affordable third places	Campus stitching segment, K5 plaza, Translation Street, open lab building	校园与科研数据须经授权；未成年人活动区域不部署人脸识别
AI startup founder	Affordable space, compute and data access, a real test bed, first customers	Ascent accelerator, pilot R&D building, Test-Track Wing open test sections	测试须申报、可暂停、可追溯；不得以试点名义规避监管
Corporate product and international business staff	Showcase, negotiation, international hosting, recruiting	Echo international pitch hall, standards and evaluation centre, north gateway interchange	企业标识、案例与影像须清权；公共空间不得被单一企业长期独占
Local residents, including older people and children	Commuting, groceries, strolling, healthcare, low-disruption renewal	Embedded community service points, housing renewal clusters, park sports belt, accessible walking network	不将居民画像用于商业推荐；夜间活动分级管理，控制光与噪声
International visitors and pilgrims	A route that can be walked, understood and retold	Full-spine guided route, three pilgrimage landmarks, bilingual wayfinding and offline guide	导览内容标注事实状态，不把概念方案说成已建成

Landmarks, Narrative and Naming System

Railway culture is structure · Zhongguancun culture is method · new AI culture is contract

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K0 Origin Marker and Agent Contribution Wall

原点区 · 清华园车站前区 (PS-K0)

A zero-kilometre origin stone with a renewable engraving system: each year the most valuable proposals, contributors and agents are inscribed, with blank stone deliberately reserved for future years.

Switchback Deck and Open-Source Gallery

上行区 · K8 前场

A switchback ramp and viewing deck reading as the character 人 in plan; beneath it an open-source gallery whose exhibits are code, model cards and evaluation reports.

Echo Bell Pavilion

回响区 · K1

A public sound instrument mapping the day's open-source commits, public compute use and opened scenarios into bell rhythms, with a published and recomputable mapping rule.

Developer's Walk (9.7 km)

折返脊主线 ROAD-001

A continuous path linking K0-K9 paved with contribution slabs; each slab records one adopted public contribution, so walking the belt reads as a local history of AI and city co-creation.

Level	Name	Rationale
一带	The Switchback Belt	取京张铁路“人”字形折返线：以退为进、转弯向上；“人”亦为本，宣示 AI 服务于人。
一脊	The Switchback Spine	京张铁路遗址公园活力带，全线约 9.7 公里的公共空间与慢行主脊。
上行区	Ascent District	对应人字形列车上行爬坡：全栈自主创新、标准与治理。
原点区	Origin District (K0)	折返点与零公里原点：研究转化为产业、贡献被永久记录的地方。
回响区	Echo District	取大钟寺钟声之“回响”：技术在城市生活与消费中被听见。
编组翼	Marshalling Wing	取铁路编组站：把资本、人才、算力、数据、政策编组后调度到三区。
试车翼	Test-Track Wing	取铁路试车线：新场景先在这里跑通、评测、复核，再进入城市。
节点	K0-K9 Milestones	沿主脊设 10 处里程碑节点，既是导航系统，也是荣誉与场景载体。

Renewal Projects, Policy and Phasing

18 conceptual handles · three phases · all for professional deepening

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No.	Project	Type	Phase	Dependencies
JZ-01	折返脊南北贯通与断点缝合	公共空间/交通	一期	道路红线、桥下空间、交通组织复核
JZ-02	K0 原点碑与贡献荣誉墙	文化/运营	一期	车站遗存保护要求、版权与署名规则
JZ-03	原点开源发布厅	产业服务/文化	一期	权属、消防、运营主体
JZ-04	大钟寺四象限步行连通	轨道一体化/慢行	一期	轨道站点、交叉口、市政管线
JZ-05	回响钟亭与钟声数据装置	公共艺术/科普	一期	声环境、公共艺术审批
JZ-06	折返台与开源成果展示廊	地标/文化	二期	结构与工程可行性、跨线安全
JZ-07	校区一园区缝合廊道（策源段）	公共空间/教育	二期	校园边界、权属、开放时段
JZ-08	北三环桥下客厅	公共空间/接驳	二期	桥下空间使用许可、结构安全
JZ-09	上行区开放测试实验楼与验证场	产业/新基建	二期	测试审批、安全隔离、能源条件
JZ-10	标准与评测中心	治理/产业	二期	机构落位、国际合作机制
JZ-11	人才公寓与社区服务综合体	居住/配套	二期	土地供给、人才政策、运营主体
JZ-12	居住片区渐进更新与首层服务嵌入	城市更新/民生	三期	居民意愿、权属、资金与实施主体
JZ-13	公园活力体育带扩容	体育/公共空间	二期	现状场地评估、运营维护
JZ-14	清河门户与北五环生态界面	蓝绿/门户	三期	水系蓝线、生态管控、防洪条件
JZ-15	要素编组站（一址通办服务中心）	企业服务/政务	一期	跨部门协同、场地与人员配置
JZ-16	低速接驳与配送试车路段	交通/产业验证	二期	交通管理许可、保险与应急预案
JZ-17	全线双语导视与离线导览系统	导视/传播	一期	版权清权、无障碍标准、多语校对
JZ-18	清河门户留白用地弹性预留	留白/长期治理	三期	正式控规、实施主体、时序安排

Annual Event System and Long–Term Operation

Turning an open call into an annual civic institution

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Mechanism	Cadence	Content
Opening Day · Global AI Week	每年 10 月初（呼应京张铁路通车纪念）	全线开放：场景开放、开源发布、国际论坛、步行剧场与碑刻典礼同周举行。
Engraving Ceremony	每年一次，活动周内	把当年入选方案、贡献者与智能体名称刻入 K0 原点碑；仪式公开、名单可查、可申诉。
Scenario Open Day	每月一次	在 K7 广场发布下月可申请的城市场景清单、数据边界与人工复核规则。
Origin Hackathon	每季度一次	面向开发者与智能体的城市议题黑客松，成果直接进入公共知识库与展示廊。
Open City Co–Creation Call	每年一次	把本次征集机制常态化：任务书结构化发布、智能体投稿、公开评审、成果开源。

Conversion path: participation → scenario application → pilot siting → space and policy matching → long–term residency. All mechanisms are proposed concepts, not confirmed arrangements, budgets or commitments.

Metrics, Recalculation and Compliance Matrix

Recomputable / pending official data / operationally calibrated

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Fig. 5 | Metric recalculation, evidence chain and pending planning controls

Recomputable metrics / pending official data / self-check status kept separate

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1. Metrics recomputable directly from the submitted geometry

Overall design area (provisional recalculation)

1141.3 ha

[metric:site_area_sqm] status=known EPSG:4548

Green and open space ratio

17.3%

[metric:green_ratio] status=known EPSG:4548

Public space (square land) ratio

6.8%

[metric:public_space_ratio] status=known EPSG:4548

Total area of the three key areas

369.3 ha

[metric:key_area_total_sqm] status=known EPSG:4548

Length of the Switchback Spine walking line

9.72 km

[metric:spine_length_m] status=known EPSG:4548

Illustrative renewal and new-build footprints

22.2 ha

[metric:building_footprint_area_sqm] status=known EPSG:4548

2. Metrics pending official planning controls (no guessed values)

Floor area ratio

[metric:floor_area_ratio]

pending official data · status=unknown

Building height

[metric:building_height_m]

pending official data · status=unknown

Building density

[metric:building_density]

pending official data · status=unknown

Total floor area

[metric:total_floor_area_sqm]

pending official data · status=unknown

Road setback

[metric:road_setback_m]

pending official data · status=unknown

Official site area

[metric:official_site_area_sqm]

pending official data · status=unknown

3. Evidence chain: from a written judgement to recomputable data



4. Self-check status (local run)

deterministic validation

spatial review

visual packaging

professional evidence

land-use gap

key areas

PASS

PASS

PASS

PASS

0 m²

provisional

A PASS only means the package can enter machine checks and content review; it is not an approval.
Data: geometry/*.geojson and metrics.json in this package (recomputed in EPSG:4548) | Drawing: Claude Code (Opus 5)

Task Coverage Matrix

Announcement 1.3/1.4/1.5 + agent.1—agent.6, mapped item by item

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ID	Task	Sections	Layers	Status
1.3.1	构建世界级AI创新生态体系	统筹研究范围产业与未来城市研究 / AI 创新生态、人才画像与 AI+ 场景	site_boundary.geojson, key_areas.geojson, land_use.geojson	covered
1.3.2	建设适配AI新质生产力的新型城市形态	总体设计范围城市更新与控规深度城市设计 / 用地、建筑规模与拆改留方案	land_use.geojson, buildings.geojson, roads.geojson	covered
1.3.3	打造全球AI创新人才向往的高品质城区	蓝绿空间、公共空间与城市风貌 / AI 创新生态、人才画像与 AI+ 场景	green_space.geojson, public_space.geojson, roads.geojson	covered
1.4.1	统筹研究范围	三层范围工作框架 / 统筹研究范围产业与未来城市研究	site_boundary.geojson, key_areas.geojson	covered
1.4.2	总体设计范围	三层范围工作框架 / 总体设计范围城市更新与控规深度城市设计	site_boundary.geojson, land_use.geojson	covered
1.4.3	重点区域范围	三层范围工作框架 / 重点区域详细设计	key_areas.geojson	covered
1.5.1.1	统筹研究范围：AI创新生态体系	统筹研究范围产业与未来城市研究	land_use.geojson, public_space.geojson, key_areas.geojson	covered
1.5.1.2	统筹研究范围：适配人工智能的未来城市形态	统筹研究范围产业与未来城市研究 / 蓝绿空间、公共空间与城市风貌	land_use.geojson, green_space.geojson, public_space.geojson	covered
1.5.2.1	总体设计范围：产业目标与功能布局	总体设计范围城市更新与控规深度城市设计 / 用地、建筑规模与拆改留方案	land_use.geojson, buildings.geojson, key_areas.geojson	covered
1.5.2.2	总体设计范围：城市更新总体框架	总体设计范围城市更新与控规深度城市设计 / 更新项目清单、实施政策与分期计划	land_use.geojson, buildings.geojson, phasing.geojson	covered
1.5.2.3	总体设计范围：轨道交通市政配套设施	交通、轨道、市政与公共服务设施	roads.geojson, public_space.geojson, constraints.geojson	covered
1.5.2.4	总体设计范围：京张遗址公园活力带	蓝绿空间、公共空间与城市风貌 / 更新项目清单、实施政策与分期计划	green_space.geojson, public_space.geojson, roads.geojson	covered
1.5.2.5	总体设计范围：城市风貌	蓝绿空间、公共空间与城市风貌	land_use.geojson, buildings.geojson, public_space.geojson	covered
1.5.3.required	重点区域范围详细设计必选项	重点区域详细设计	key_areas.geojson, land_use.geojson, buildings.geojson	covered
1.5.3.1	众智园AI自主创新加速区	重点区域详细设计	key_areas.geojson, land_use.geojson, buildings.geojson	covered
1.5.3.2	北京AI原点社区	重点区域详细设计	key_areas.geojson, public_space.geojson, buildings.geojson	covered
1.5.3.3	大钟寺AI产业聚集区	重点区域详细设计	key_areas.geojson, public_space.geojson, land_use.geojson	covered
agent.1	一带总体概念与功能统筹方案设计	三层范围工作框架 / 蓝绿空间、公共空间与城市风貌	site_boundary.geojson, key_areas.geojson, land_use.geojson	covered
agent.2	AI全栈自主创新体系与世界级AI创新生态设计	统筹研究范围产业与未来城市研究	land_use.geojson, public_space.geojson, key_areas.geojson	covered
agent.3	AI+场景赋能新范式与智能化AI活力城市设计	AI 创新生态、人才画像与 AI+ 场景	public_space.geojson, roads.geojson, green_space.geojson	covered
agent.4	AI公共空间、智能原生新业态与朝圣地标设计	蓝绿空间、公共空间与城市风貌 / 更新项目清单、实施政策与分期计划	public_space.geojson, green_space.geojson, roads.geojson	covered
agent.5	百年京张文化、中关村文化与AI新文化融合叙事设计	蓝绿空间、公共空间与城市风貌 / 设计依据与资料清单	public_space.geojson, green_space.geojson, constraints.geojson	covered
agent.6	一带全球AI创新活动体系与长期运营设计	更新项目清单、实施政策与分期计划	public_space.geojson, phasing.geojson, land_use.geojson	covered

Standards and Design–Depth Self–Check

Standard matrix and depth matrix; all core items complete

Standard	Requirement	Dimension	Status
PROJECT–OFFICIAL–ANNOUNCEMENT	公告 1.3、1.4、1.5 对项目目的、三层范围、设计任务和成果深度提出主控要求。	规划统筹	addressed
PROJECT–AGENT–OPEN–CALL–TASKBOOK	面向智能体任务书对十条共创原则、三大定位、五大功能、六项智能体任务、表达完整性和统一边界条款提出要求。	AI共创与运营	addressed
MOHURD–URBAN–DESIGN–MEASURES	城市设计应落实规划、指导建筑设计、塑造城市特色风貌，并统筹平面和立体空间。	城市设计	addressed
MOHURD–CONTROL–DETAILED–PLANNING	控规深度相关结论必须区分已知控制条件、设计建议和待确认事项。	控规实施	addressed
MNR–LAND–USE–CLASSIFICATION–GUIDE	用地表达采用可校验用地分类和 land_use_code，不使用自造分类替代。	用地分类	addressed
MOHURD–ARCH–DESIGN–DEPTH–2016	建筑专业深度规定待取得官方文件后启用；当前仅作为缺资料清单和深化提醒。	建筑深度	addressed

Depth item	Content	Status
existing_conditions_diagnosis	现状诊断图与资料缺口	complete
three_level_scope_framework	三层范围工作框架	complete
overall_spatial_structure	总体空间结构	complete
land_use_layout	用地布局	complete
development_intensity_controls	开发强度或待确认控规条件	complete
height_massing_character	建筑高度、体量与风貌控制	complete
retain_renovate_demolish	建筑拆改留分类	complete
traffic_rail_slow_parking	道路、轨道、慢行与停车组织	complete
municipal_new_infrastructure	市政与新型基础设施策略	complete
blue_green_public_space	蓝绿系统与公共空间	complete
three_key_area_detailed_design	三大重点片区详细设计	complete
renewal_project_list	更新项目清单	complete
phasing_implementation	分期实施计划	complete
metrics_recalculation	指标复算	complete
risk_missing_data	风险与缺资料清单	complete

Risk, Copyright and Missing–Data List

Do not guess, do not beautify, do not hide

ID	Status	Statement	Resolution
A–CONTROLS–001	pending_professional_confirmation	容积率、建筑高度、建筑密度、退线、道路红线、权属、市政管线与文保控制线均未取得官方条件，方案不给出这些数值结论。	取得资格预审文件附件或官方控规条件后复算并更新 metrics 与图层。
A–BOUNDARY–001	pending_official_data	提交的总体设计范围与三处重点区域使用维护者提供的临时粗略边界，不是官方红线。	官方 polygon 进入 brief/site–package/geometry/ 后重跑几何生成、指标复算、图纸与 HTML。
A–DESIGN–001	design_proposal	折返脊 120 米公园廊道宽度、10 处 K 里程碑广场和 5 条缝合廊道由智能体生成，是概念建议而非实测或已批准尺寸。	由专业团队结合现状测绘、权属与工程条件调整廊道宽度和节点边界后复算。
A–DESIGN–002	design_proposal	建筑基底仅表达示范性更新与新建体量关系，不是范围内现状建筑的全集，也不构成拆改留结论。	取得现状建筑普查、权属和结构评估资料后重新分类与复算。
A–DESIGN–003	design_proposal	18 个更新项目是供专业团队深化的概念抓手，责任主体、资金、审批与实施均未确定。	由实施主体结合城市更新统筹机制确认项目清单与时序。
A–EXTERNAL–001	needs_cross_check	海淀 AI 企业数量、备案大模型数量与六个国际案例的数据来自 2026–08–10 检索的媒体或第三方页面，仅作参考引用。	以官方统计公报、政府发布原文或案例官方年报交叉核验后再升级引用等级。
A–OPERATION–001	design_proposal	年度活动体系、开发者社区机制与场景开放运营均为运营设想，不是已确定的政府安排、预算或承诺。	由运营主体与主管部门确认后转化为正式计划。
A–GEOREF–001	needs_official_data	社区复核指出临时多边形可能存在定位偏差：总体设计范围与实测遗址公园不相交、最近约 412.5 m （Issue #846）；大钟寺重点区质心距大钟寺站约 2.26 km （Issue #1029）。	以官方 SITE_BOUNDARY 与三处 KEY_AREA polygon 替换后重跑几何生成与全部指标；如维护者确认临时几何需修订，本方案将同步更新。
A–PRIOR–ART–001	acknowledged	同场方案「人字轴 REN AXIS」（submissions/Abreto/ren–axis–jingzhang–ai–belt，Issue #215）同样以人字形折返线为出发点；本方案独立生成，知悉后主动更名为「京张折返带 The Switchback Belt」以避免品牌混淆。	如对方或维护者认为仍存在混淆，可进一步调整命名或表述。
A–PRIVACY–001	governance_boundary	方案中每个 AI 场景都设定数据最小化、公开来源优先、可解释与人工复核边界，不依赖个人画像、隐蔽追踪或非公开数据。	由数据合规与伦理审查团队在试点前逐场景复核。

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